

PRAMAN

Personal High-Selectivity Trading System

Refined Architecture, Development Roadmap and Agent Instruction Pack

Objective: build a selective, auditable system that may take zero to a few trades when evidence, calibrated probability, expected value and risk controls align.

Important boundary

The objective is not to guarantee profit or force daily trading. The system must be designed to abstain. Profitability must be demonstrated by point-in-time historical testing, forward paper trading and small-capital live validation before automation is trusted.

Prepared as an instruction-ready specification for coding agents

How to use this document

- Give the relevant phase section to Codex/Claude/Antigravity. Do not ask an agent to implement the entire document in one session.
- Treat the existing repository, frozen pre-registration, AGENTS.md/CLAUDE.md, HANDOFF.md and tests as the source of truth.
- Implement one bounded feature at a time, add tests, run the full suite at the phase boundary, update HANDOFF.md, commit and push.
- Do not modify protected historical/forward experiments after seeing outcomes. New trading research should be registered separately.

Section	Purpose
1. Product definition	What Praman should become and what it must not become
2. Target architecture	The full trading pipeline
3. Decision contract	Exact outputs and abstention logic
4. Intelligence modules	Market, sector, event, fundamental, analogue and model layers
5. Alpha and confidence	Labels, models, calibration, uncertainty and meta-selection
6. Risk / portfolio / execution	Deterministic safety layer and realistic fills
7. Evaluation	How to prove value without fooling ourselves
8. Automation levels	Paper -> assisted -> small live -> automated
9. Build roadmap	Fastest sensible implementation order
10. Acceptance criteria	Definition of done for each stage
11. Master agent prompt	Copy-paste instruction for coding agents

1. Product definition

One-line definition

Praman should become an evidence-constrained, high-selectivity personal trading decision system for NSE equities that scans broadly, trades rarely, quantifies uncertainty and expected value, enforces deterministic risk, and defaults to NO TRADE when evidence is insufficient.

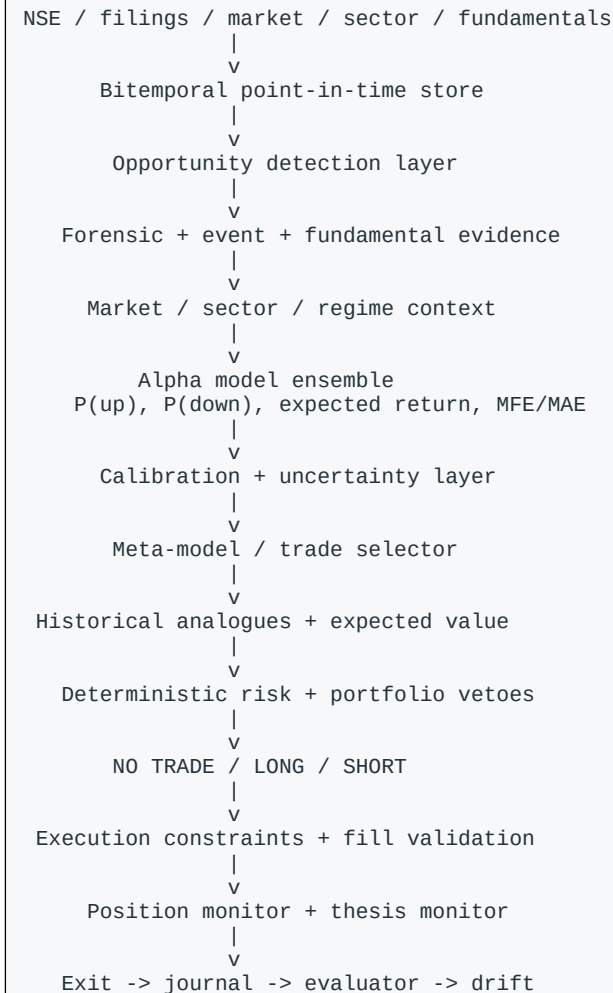
Current Praman vs. trading Praman

Current forensic layer	New trading layer
Explains unusual price moves	Estimates whether a tradable opportunity exists
Measures disclosures, delivery, volume, surveillance and subsequent behaviour	Combines market/sector context, alpha models, event/fundamental evidence, analogues and portfolio constraints
Avoids manipulation conclusions	Avoids profit guarantees and unsupported confidence claims
Deterministic, point-in-time and auditable	Keeps the same audit discipline while adding trading-specific targets and execution
Research output	NO TRADE / LONG CANDIDATE / SHORT CANDIDATE, followed by risk and execution checks

Core design principles

- Abstention first: zero trades is a valid and often desirable outcome.
- Probability is not enough: require positive expected value after costs and acceptable downside.
- Calibrated confidence: a reported 70% must mean approximately 70% on comparable unseen cases.
- Point-in-time or nothing: no future knowledge in features, labels, analogues, fundamentals or disclosures.
- AI for interpretation; deterministic code for calculations, gates, PnL, risk, sizing and execution constraints.
- Separate long and short research; do not treat low long probability as a short signal.
- Every decision must be reproducible from data snapshot, code commit, rulebook and model version.
- No automatic capital deployment until paper/live readiness gates are satisfied.

2. Target architecture



Separation of responsibilities

Layer	Allowed	Not allowed
LLM / language layer	Read disclosures, extract structured events, summarize evidence, generate bull/bear/skeptic arguments	Calculate PnL, size positions, change risk limits, decide final execution
ML / statistical layer	Estimate directional probabilities, expected returns, adverse/favorable excursion, meta-selection	Override hard risk or data-quality vetoes
Deterministic engine	Costs, sizing, risk budgets, portfolio caps, data freshness, execution checks, journal, replay	Invent evidence or infer unavailable facts
Human / policy	Approve risk policy, decide automation level, review ambiguous cases	Change frozen research after seeing protected outcomes

3. Decision contract

The system should not output a naked BUY/SELL score

Every candidate must expose evidence quality, calibrated probability, uncertainty, expected value, downside, liquidity, portfolio effect and the reason it is accepted or rejected.

Candidate output schema

symbol
direction: LONG SHORT NONE
horizon: 1d 3d 5d 10d
state: NO_TRADE WATCH ELIGIBLE VETO
calibrated_probability
probability_interval / uncertainty_score
expected_net_return_after_costs
expected_loss_if_wrong
expected_value
maximum_adverse_excursion_estimate
maximum_favorable_excursion_estimate
historical_analogue_count
market_regime
sector_regime
evidence_completeness
forensic_flags
liquidity_state
portfolio_incremental_risk
invalidations[]
rejection_reasons[]
model_version / rulebook_version / data_as_of / commit_hash

Final state logic

State	Meaning
NO_TRADE	No candidate has enough evidence/edge; normal system output.
WATCH	Interesting but one or more required conditions are not yet satisfied.
ELIGIBLE	All research and deterministic risk conditions are satisfied; still subject to execution checks.
VETO	Hard data, risk, liquidity, surveillance, portfolio, execution or operational rule blocks the trade.

Critical rule

Do not impose a daily trade quota. The scanner may inspect thousands of stock-days and produce zero eligible trades.

4. Intelligence modules to add

Module	Purpose
Market intelligence	Nifty/equal-weight trend, breadth, volatility, dispersion, liquidity and regime. A stock signal is interpreted within market conditions.
Sector intelligence	Stock-vs-sector and sector-vs-market relative strength; sector breadth and volatility; prevents treating a broad sector move as stock-specific alpha.
Opportunity detector	Scans the whole eligible universe for unusual price/volume/delivery, relative-strength changes, disclosures and other preregistered triggers.
Event intelligence	Parses exchange announcements and classifies event type, direction, magnitude, novelty, certainty, effective date and source.
Minimal fundamentals	Revenue, profit, margins, cash flow, debt, coverage, ROE/ROCE, promoter holding/pledge and valuation using point-in-time availability.
Expectation vs. reality	Normalizes event size by company scale and, where legitimate data exists, compares reported outcomes with prior expectations.
Historical analogue engine	Finds genuinely comparable historical setups and reports base rates, return distribution, MFE, MAE and target-before-stop frequencies.
Bull/Bear/Skeptic/Pre-mortem	Structured challenge layer that identifies supporting evidence, contradictions, shared assumptions and plausible failure paths.
Portfolio intelligence	Correlation, sector concentration, beta/factor concentration, liquidity concentration, stress loss and effective number of independent bets.
Drift monitor	Tracks calibration, feature distributions, expectancy, slippage and performance by regime; can automatically downgrade models to paper-only.

5. Alpha, confidence and trade selection

5.1 Trading-specific labels

- Preserve the frozen forensic/forward experiment. Create a separate trading research registry and separate labels.
- Evaluate multiple horizons: 1, 3, 5 and 10 sessions. Do not assume one horizon works for every signal family.
- Prefer path-aware labels (target-before-stop / stop-before-target / timeout) over only terminal return.
- Use volatility-adjusted barriers and account for executable next-session prices and costs.
- Generate LONG and SHORT labels separately and respect real-world execution constraints.

5.2 Model stack

Component	Recommended starting point	Reason
Baseline	Logistic regression	Interpretable benchmark; exposes whether complexity is actually needed.
Tree model	LightGBM or XGBoost	Strong for nonlinear tabular interactions and missingness.
Alternative	Random Forest / ExtraTrees	Diversity for ensemble disagreement.
Sequence model	Only after tabular baselines	Use only if walk-forward evidence shows incremental value.
LLM	Not an alpha calculator	Use for structured reading/extraction, not return prediction.

5.3 Calibration and uncertainty

- Calibrate probabilities out-of-sample using Platt/logistic scaling or isotonic regression as appropriate.
- Report reliability diagrams and Brier score; do not call raw model scores probabilities unless calibrated.
- Quantify uncertainty with bootstrap/model dispersion/conformal or other preregistered methods appropriate to the model.
- Require both sufficiently high calibrated probability and sufficiently low uncertainty.

5.4 Meta-selection

Recommended structure

Stage A predicts direction/return characteristics. Stage B predicts whether Stage A should be trusted enough to trade. This allows the system to reject most model predictions.

```
Broad universe
-> trigger / opportunity
-> directional models
-> calibration
    -> meta-selector
        -> evidence / analogue checks
            -> risk + portfolio
                -> 0 to few eligible trades
```

5.5 Expected value, not win rate

Core decision quantity: $EV = P(\text{win}) \times \text{average win} - P(\text{loss}) \times \text{average loss} - \text{costs}$.

- A high win rate can still lose money if losses are much larger than wins.
- Require positive expected value after brokerage, taxes/fees, slippage assumptions and realistic fill logic.

6. Risk, portfolio and execution

6.1 Deterministic risk stack

Gate	Examples
Data validity	Freshness, as-of correctness, corporate actions, identity mapping, missing-data thresholds
Security risk	Liquidity, price bands/dynamic ranges, surveillance categories, gap risk
Trade risk	Planned stop, stress loss, quantity, maximum position size, risk/reward
Portfolio risk	Open-risk budget, sector concentration, correlation clusters, beta/factor concentration
Behavior/operations	Drawdown brake, losing-streak brake, open high-severity defects, stale ingestion, backup/restore health
Execution	Maximum acceptable gap/slippage, actual fill risk recheck, no-fill/suspension handling

6.2 Execution policy

- Decision price and research evidence are frozen at decision time.
- Execution is a separate append-only event using an actually available future price.
- Before any live order, apply a maximum acceptable entry/slippage policy. If the opportunity deteriorates before fill, cancel rather than chase.
- After fill, recompute actual risk and portfolio impact. Never silently resize or change a frozen historical experiment.
- Short execution must have its own adapter for actual instrument availability and applicable exchange/broker rules.

6.3 Kill switches

Condition	Required action
Stale or inconsistent market data	NO NEW TRADES
Risk state unavailable	NO NEW TRADES
Daily/weekly drawdown limit breached	Freeze new entries; manage exits only
Repeated order/execution failures	Disable automated execution
Model calibration/edge degradation	Reduce size or revert model to PAPER ONLY
Critical unresolved defect	Operational gate fails

7. How to prove the system adds value

Primary research objective

Do not optimize for the best-looking backtest. Optimize for evidence that survives unseen time periods, realistic costs, parameter perturbations and regime changes.

Evaluation ladder

```
Development period
-> walk-forward / purged validation
-> untouched holdout
-> forward shadow opportunities
-> paper trading
-> small live capital
-> scale only after edge-confidence gate
```

Audit dimension	Minimum evidence
Discrimination	AUC/PR metrics where appropriate, but never as the only trading metric
Calibration	Reliability curve, Brier score, observed win rate by probability bucket
Economics	Net expectancy after costs, profit factor, payoff ratio, benchmark-relative return
Tail risk	MAE, severe-loss frequency, gap-through-stop, circuit/illiquidity stress where data permits
Stability	Performance by year/regime/sector/cap bucket; parameter-neighbour stability
Uncertainty	Bootstrap confidence intervals and explicit sample counts
Multiple testing	Record every attempted strategy/parameter family; apply appropriate correction/deflated metrics
Process quality	Good-loss/bad-win matrix, rule adherence, overrides, missing exit provenance
Execution realism	Actual next-available prices, slippage/cost assumptions, no-fill cases

No-retrospective-tuning rule

- Once a registered experiment has been evaluated, do not alter that experiment and call the altered version the same test.
- New hypotheses are allowed, but they must become a new strategy/version with their own untouched evaluation window.
- Keep failed strategies visible in the strategy registry.

8. Automation levels

Level	Mode	Requirements
A0	Research only	Historical scanner/evaluator; no order creation.
A1	Paper trading	Automated candidate generation and paper fills; human reviews all decisions.
A2	Assisted live	System proposes trades and exact risk; human explicitly confirms each order.
A3	Small-capital constrained auto	Only preapproved strategies, small size, deterministic hard gates and kill switches; human monitors.
A4	Broader automation	Only after substantial live evidence, stable calibration/expectancy and operational reliability.

Recommended current destination

Build toward A1 first, then A2. Do not jump directly from historical research to A3/A4.

9. Fastest sensible build roadmap

Phase	Scope	Exit condition
T0 - Freeze & baseline	Close existing replay/evaluator defects, verified backups, clean handoff	Full suite green; current experiment frozen and reproducible
T1 - Market context	Market/sector regime, breadth, volatility, dispersion, relative strength	Point-in-time features + tests + historical coverage report
T2 - Trading dataset	1/3/5/10-day long/short path-aware labels and opportunity dataset	No leakage; registered label definitions; reproducible dataset
T3 - Baseline alpha	Logistic + tree models, walk-forward evaluation	Baselines documented, costs included, no tuning on final holdout
T4 - Calibration/meta	Calibrated probabilities, uncertainty, meta-selector, abstention	Calibration report and selectivity/coverage curves
T5 - Analogues/events	Historical analogue engine + structured disclosure/event extraction	Evidence provenance and validation benchmark
T6 - Fundamentals	Minimal PIT fundamentals + event materiality by company scale	Coverage/freshness thresholds documented
T7 - Portfolio/execution	Correlation/risk clusters, fill policy, slippage, short constraints	Paper engine rejects invalid fills and portfolio breaches
T8 - Paper forward	Automated daily scan + paper trading + evaluator + drift	Sufficient duration/trades/opportunities; no serious rule violations
T9 - Assisted live	Human-confirmed small-capital execution	Operational gate passed; live monitoring/kill switches verified
T10 - Limited automation	Only proven strategies and predefined risk envelope	Edge-confidence gate passed with live/forward evidence

Parallel work allowed

A read-only web cockpit may be built alongside T1-T7, but it must consume backend outputs and must not reimplement calculations in JavaScript.

10. Definition of done / acceptance criteria

Area	Acceptance condition
Data	Every trading feature has event date + knowledge date; PIT query tests include adversarial future-data cases.
Labels	Trading labels are preregistered before evaluation; path ordering and global-session handling are tested.
Models	Every complex model beats a transparent baseline on untouched data or is rejected.
Calibration	Probability buckets match observed frequencies within documented uncertainty; sample counts shown.
Selectivity	Performance is reported as a function of coverage (e.g., top 1%, 5%, 10% candidates), not only a single threshold.
Costs	All headline trading metrics are after realistic fees/costs and a documented slippage model.
Risk	A candidate cannot bypass deterministic security/trade/portfolio/operational gates.
Execution	Decision, order request, fill and post-fill risk are separate append-only events.
Reproducibility	Decision can be replayed using commit, data snapshot/hash, rulebook, model and prompt versions.
Paper/live	Automation level cannot increase until the relevant readiness gate passes.
Monitoring	Drift/calibration/expectancy degradation can automatically downgrade a strategy to paper-only.
UI	Dashboard is a view/controller over backend truth; no duplicate risk/PnL logic in frontend.

11. Copy-paste master instruction for coding agents

How to use

Do not give this whole project to one agent in one run. Paste this master instruction, then append exactly one phase/task from Section 9.

You are working on PRAMAN, an existing NSE forensic/research repository that is being extended into a high-selectivity personal trading decision system.

SOURCE OF TRUTH

1. Verify the repository root and git status.
2. Read AGENTS.md / CLAUDE.md, docs/HANDOFF.md, relevant DESIGN/rulebook docs, and only the files needed for the assigned task.
3. Existing frozen pre-registrations and protected forward windows MUST NOT be altered or evaluated early.
4. Preserve unrelated uncommitted work.

PRODUCT OBJECTIVE

Build an auditable system that scans broadly but may take zero to a few trades. A trade is eligible only when point-in-time evidence, calibrated probability, expected value after costs, uncertainty, deterministic risk, liquidity and portfolio constraints align. NO TRADE is a first-class successful output.

NON-NEGOTIABLE ARCHITECTURE

- Keep the existing forensic classifier and frozen research separate from new trading research.
- Every feature/fact must be point-in-time: event date + knowledge date + as-of reads.
- LLMs may read/extract/summarize evidence; they may not calculate PnL, position size, risk limits, portfolio mutation or execution decisions.
- LONG and SHORT research are separate.
- Raw model scores are not probabilities until calibrated.
- Optimize expected value after costs and downside, not win rate alone.
- Deterministic risk/portfolio/operational vetoes override ML/LLM output.
- Decision, execution/fill, monitoring and exit are separate append-only events.
- Never force a daily trade quota.
- Do not add broker automation in the current task unless explicitly authorized.

ENGINEERING PROCESS

- Before edits: report baseline git status and targeted/full test state as appropriate.
- Write a short implementation plan and identify invariants affected.
- Implement only the assigned phase/task.
- Add adversarial tests, especially for PIT leakage, corporate actions, identity mapping, costs, execution ordering and risk boundaries.
- Use temporary/read-only copies for proofs and experiments; never clear/recreate durable stores.
- Run targeted tests during development and the full suite once at the task/phase boundary.
- Update docs/HANDOFF.md with exact completed work, limitations, unresolved defects and next step.
- Commit coherent tested changes. Push when requested/available.

RESEARCH PROCESS

- Register label/metric/threshold definitions before observing evaluation results.
- Preserve all failed strategies and parameter attempts in the strategy registry.
- Report sample counts, missingness, confidence intervals and regime/year breakdowns.
- Never tune a frozen experiment after seeing its outcome. New ideas require a new version and new evaluation.

FINAL REPORT

Return: changed files, tests, commit(s), data/research artifacts created, PASS/PARTIAL/MISSING against the assigned acceptance criteria, unresolved defects/limitations, and the safest next task.

12. Per-phase prompt template

Continue PRAMAN from the current committed HANDOFF.

ASSIGNED PHASE: [insert exactly one phase/task]

Goal:

[one clear outcome]

Required deliverables:

- [module / schema / command / report]
- tests for [critical invariants]
- documentation / HANDOFF update

Acceptance criteria:

- [copy only the relevant criteria from this blueprint]

Do not:

- modify frozen preregistration or protected outcomes
- add unrelated features
- change risk limits just to improve historical performance
- rewrite deterministic calculations in the UI/LLM
- start broker automation

Work in small checkpoints. Run targeted tests during implementation and the complete suite once before the final commit. Report PASS/PARTIAL/MISSING against every acceptance criterion.

Recommended next instruction

Next phase after current closeout

T1 - Market and sector context: build point-in-time market/sector regime, breadth, volatility, dispersion and stock-vs-sector/market relative-strength features. Do not yet train the final trading model.

13. What the final user should see

PRAMAN - DAILY DECISION DESK

Universe scanned: 2,695
 Candidates: 34
 Passed evidence/data checks: 11
 High-confidence model candidates: 4
 Passed risk/portfolio/execution checks: 2

Candidate ABC
 Direction: LONG
 Calibrated P(profitable): 0.xx
 Uncertainty: LOW / MEDIUM / HIGH
 Expected net return: x.x%
 Expected loss if wrong: x.x%
 Expected value: +x.xxR
 Historical analogues: N = xxx
 Market: supportive / neutral / hostile
 Sector: supportive / neutral / hostile
 Evidence completeness: xx%
 Risk: PASS
 Portfolio: PASS
 Execution: WAIT / ELIGIBLE / CANCEL
 Final state: WATCH / ELIGIBLE / VETO

Most important output on many days:
 NO TRADE - insufficient expected value or confidence.

Success criterion for the product

Praman is successful when it can reliably explain why it is trading, why it is not trading, how uncertain it is, how much can be lost, and whether its historical/forward evidence actually supports the decision. A visually impressive dashboard or high model score is not success by itself.

END OF BLUEPRINT